

Capítulo 9

Chapter Precap

- Introducir predictores cuantitativos para modelar la variación a lo largo de líneas.
- Primero, se presenta un modelo con una sola pendiente e intersección.
- A continuación, se introducen modelos con un único predictor cuantitativo y un único predictor categórico, pero sin interacción.
- Describimos modelos con una interacción entre un predictor cuantitativo y un predictor categórico, que pueden representar interceptos y pendientes específicas de grupo.
- Presentamos un modelo de efectos aleatorios que incluye un único predictor cuantitativo pero con "pendientes aleatorias" para cada participante.
- Por último, se presenta un modelo más amplio en el que la variación tanto en las intersecciones como en las pendientes de las líneas se predice utilizando dos predictores categóricos y su interacción, además de incluir efectos "aleatorios" dependientes del oyente para todos los predictores y su interacción.

Data

```
library (brms)
library (bmmb)

data (exp_data)

options (contrasts = c('contr.sum', 'contr.sum'))

# make a copy of the VTL predictor and center it
→ exp_data$vtl_original = exp_data$vtl
→ exp_data$vtl = exp_data$vtl - mean(exp_data$vtl)
```

- **height**: A number representing the *height* (in centimeters) reported for the speaker on each trial.
- • **vtl**: An estimate of the speaker's *vocal-tract length* (VTL) in centimeters.
- **C**: A letter representing the speaker *category* (**b**=boy, **g**=girl, **m**=man, **w**=woman) reported by the listener for each trial.
- **G**: The *apparent gender* of the speaker indicated by the listener, **f** (female) or **m** (male).
- **A**: The *apparent age* of the speaker indicated by the listener, **a** (adult) or **c** (child).

Research Questions

- (Q1) Is there a linear relationship between speaker VTL and apparent height?
- (Q2) Can we predict expected apparent height responses given only knowledge of speaker VTL?

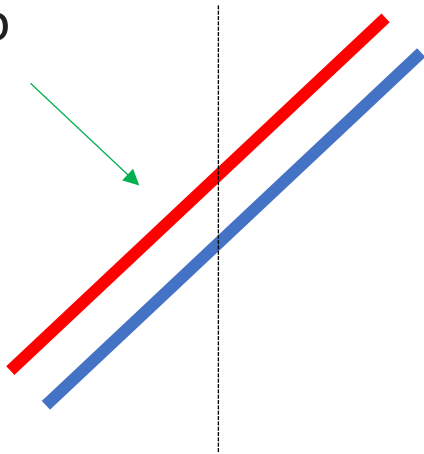
- Por primera vez incluimos :
 - Un predictor cuantitativo.
 - Variación del modelo a lo largo de las líneas.

Lines

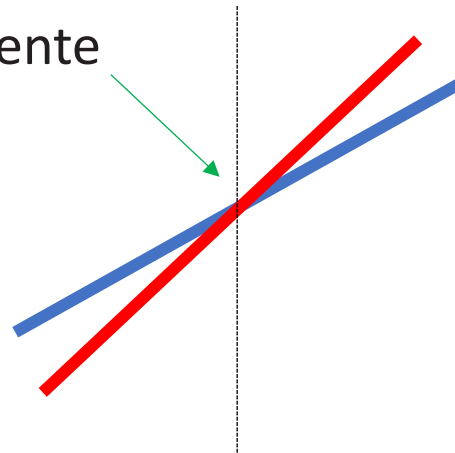
- Las líneas tienen dos parámetros:
 - El intercepto determina el valor de la línea donde $x = 0$.
 - Una pendiente determina cuánto se eleva la recta por cada cambio de unidad en x .

$$y = a + b \cdot x$$
$$\mu = \text{Intercept} + \text{slope} \cdot x$$

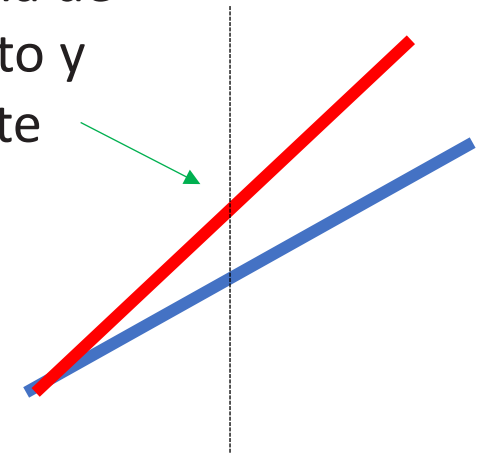

Diferencia de intercepto



Diferencia de pendiente



Diferencia de intercepto y pendiente



Secret Lines

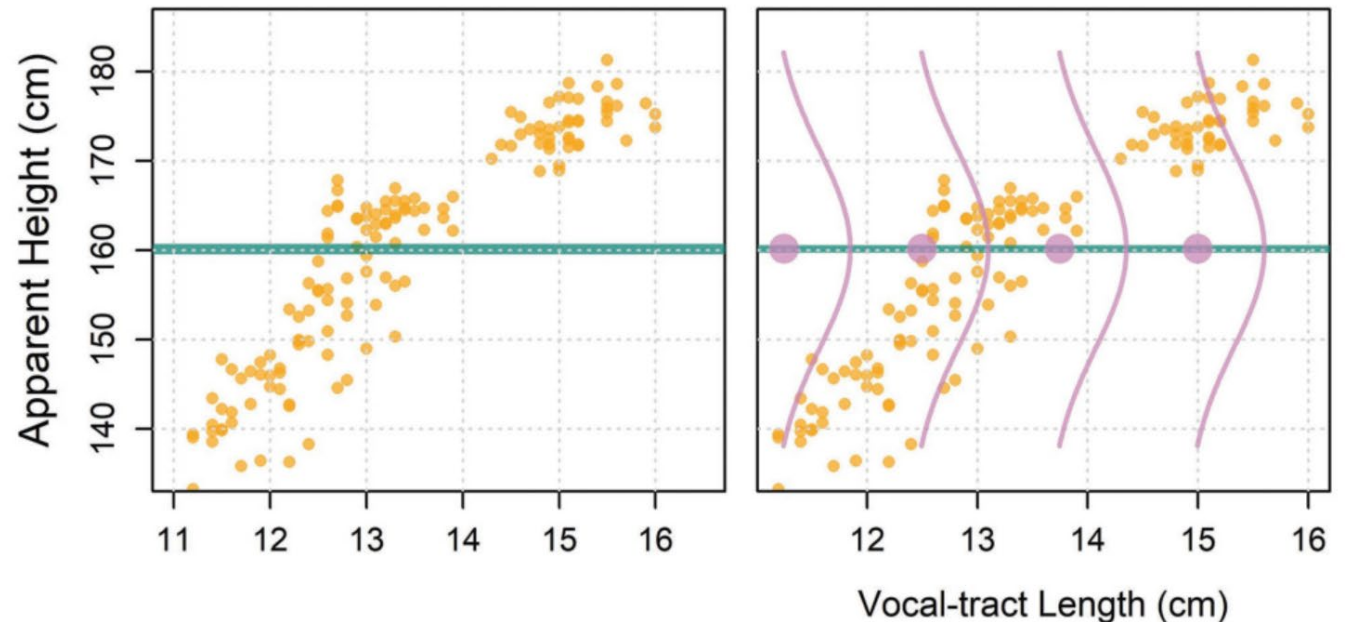
- Hemos estado modelando "secretamente" la variación a lo largo de las líneas todo este tiempo.
- Hemos estado fijando todas las pendientes en 0.
- ¿Hay una pendiente "mejor" para nuestros datos?

Intercept = 160, slope = 0

$$\mu = \text{Intercept} + \text{slope} \cdot \text{vtl}$$

$$\mu = \text{Intercept} + 0 \cdot \text{vtl}$$

$$\mu = \text{Intercept}$$



The 'Best' Line

- ¿Qué significa "mejor"?
- En general*, la "mejor" pendiente es la que minimiza los residuos: la distancia de los puntos a la recta.

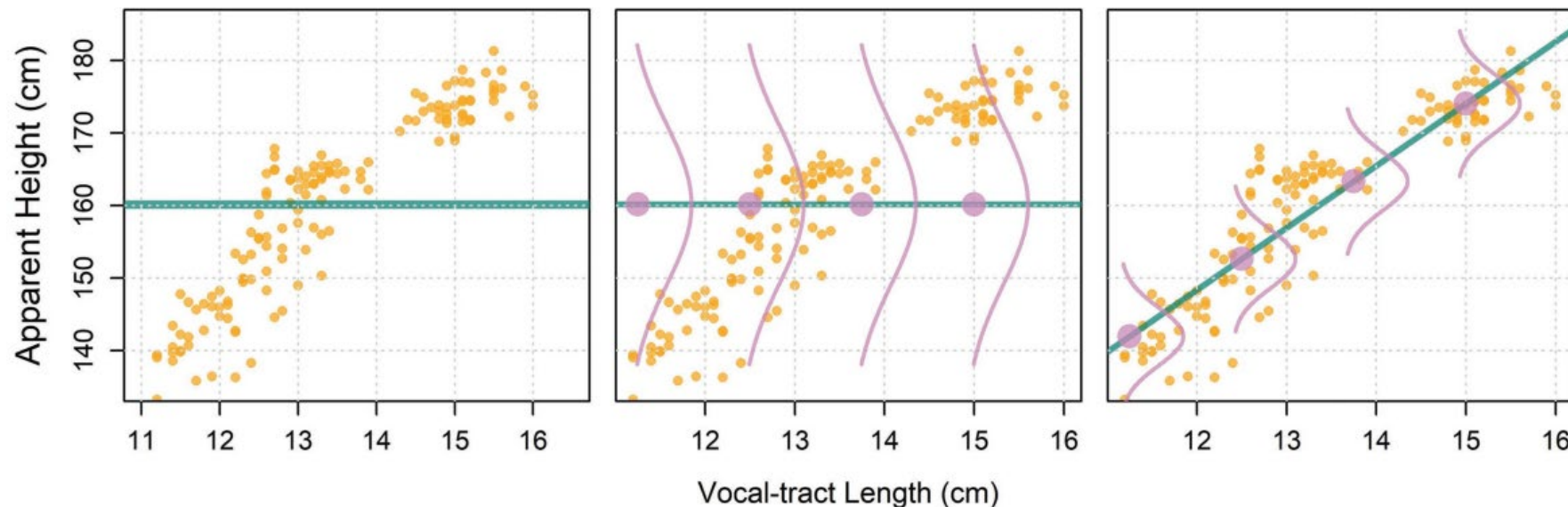
$$r[i] = \left(\text{Intercept} + \text{slope} \cdot \text{vtl}[i] \right) - \text{height}[i]$$

- Minimizar los residuales significa maximizar la probabilidad, es decir, hacer que los datos sean más probables según nuestro modelo.

*Para los modelos multinivel, la "mejor" línea no será necesariamente la que minimice los residuos.

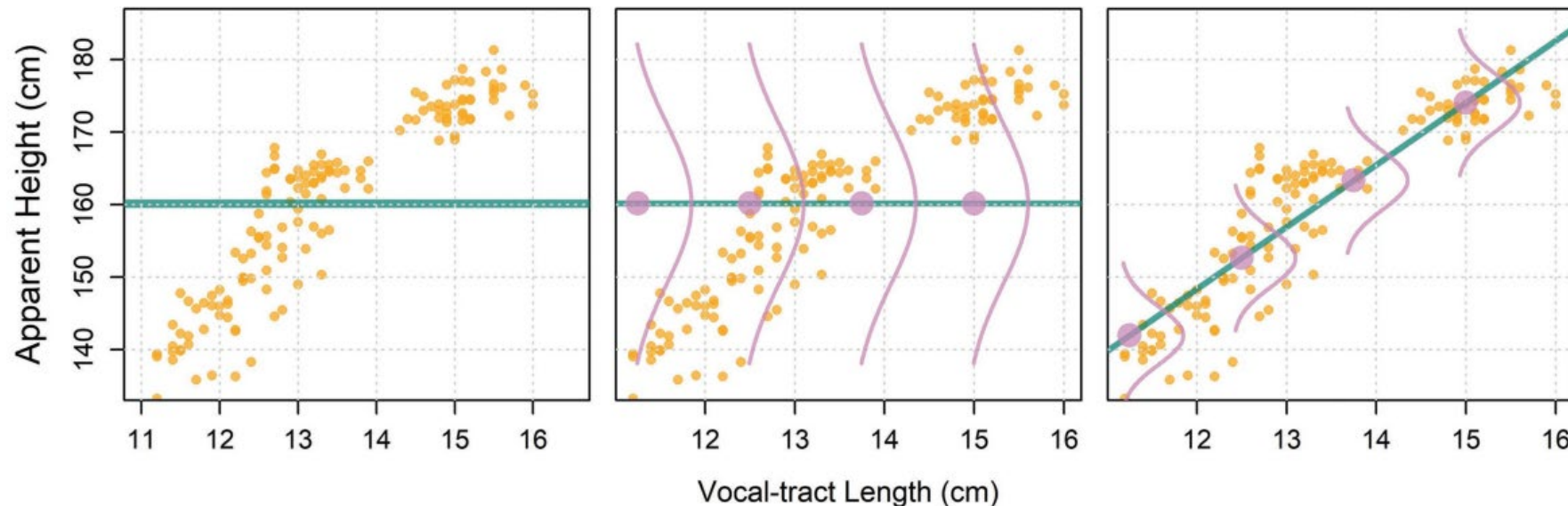
Minimizing Residuals

- Una pendiente de cero conduce a una gran varianza de error (eso es malo).
- La pendiente distinta de cero hace que nuestros residuales y la varianza del error sean mucho más pequeños (eso es bueno).



Variation along Lines

- Antes, nuestra distribución normal se encontraba en un solo lugar, generando datos.
- Ahora nuestra distribución se desliza generando datos basados en el valor de x.



Description of the Model

- Comenzaremos con 'modelos uninivel de efectos fijos' solo para comprender la geometría de estos modelos.

```
height ~ vtl
```

$$\text{height}_{[i]} \sim N(\mu_{[i]}, \sigma)$$

$$\mu_{[i]} = \text{Intercept} + VTL \cdot \text{vtl}_{[i]}$$

Parámetro de
pendiente

Priors:

$$\text{Intercept} \sim t(3, 160, 12)$$

$$VTL \sim t(3, 0, 12)$$

$$\sigma \sim t(3, 0, 12)$$

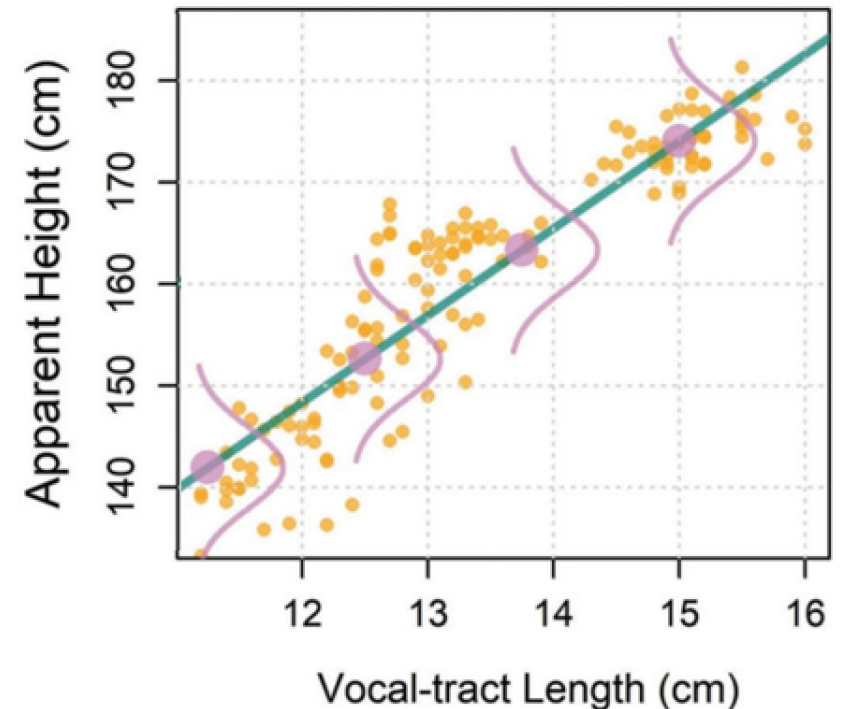
predictor

Description of the Model

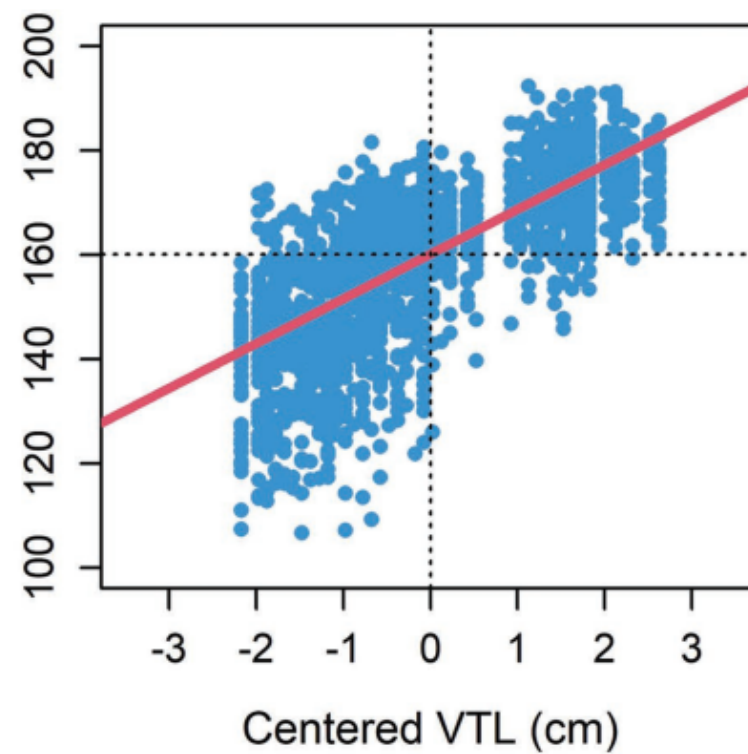
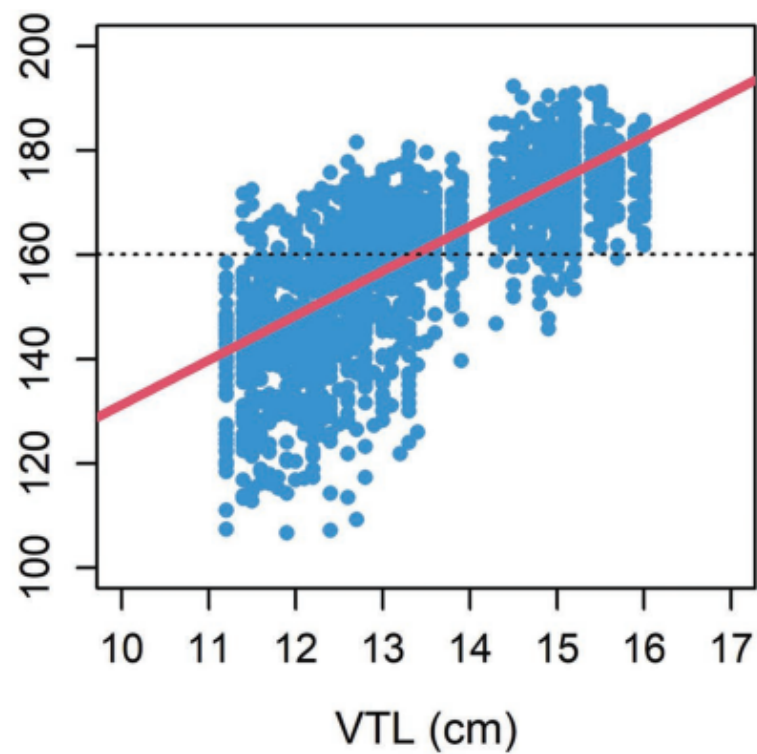
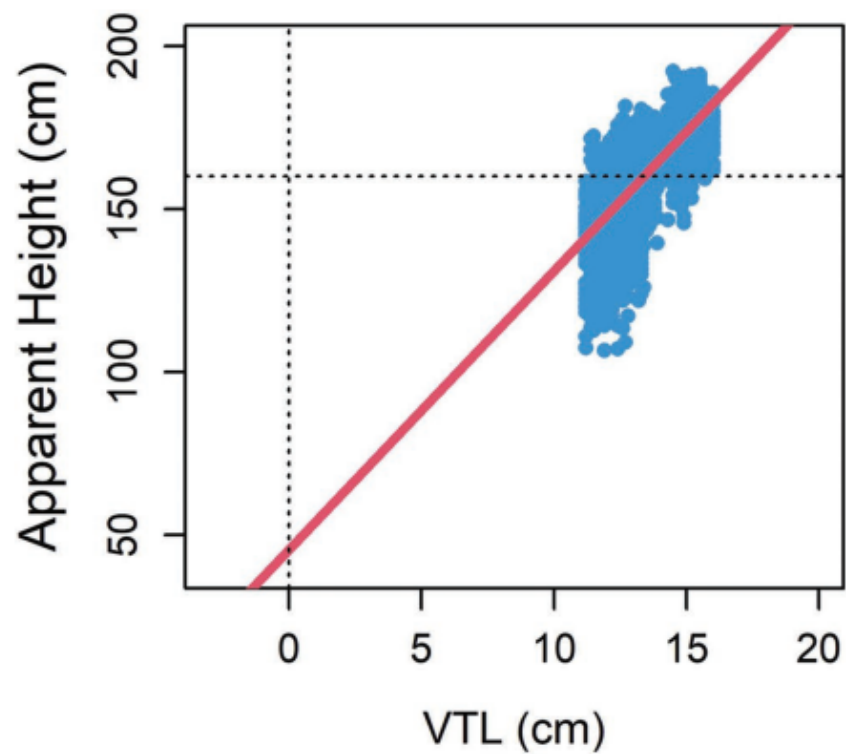
- Dos maneras de pensar en este modelo.

$$\text{height}_{[i]} = \text{Intercept} + VTL \cdot \text{vtl}_{[i]} + N(0, \sigma)$$

$$\text{height}_{[i]} = N(\text{Intercept} + VTL \cdot \text{vtl}_{[i]}, \sigma)$$



Centering Predictors



Centering

- Si 0 no es significativo para el predictor, centrarlo puede:
 - Hacer que la intersección represente un valor más significativo.
 - Facilitar la especificación de prioridades.
 - Facilitar la comparación de los efectos principales en presencia de interacciones en algunos casos.
- Recomendación: Si 0 no es un valor significativo para el predictor, céntralo.

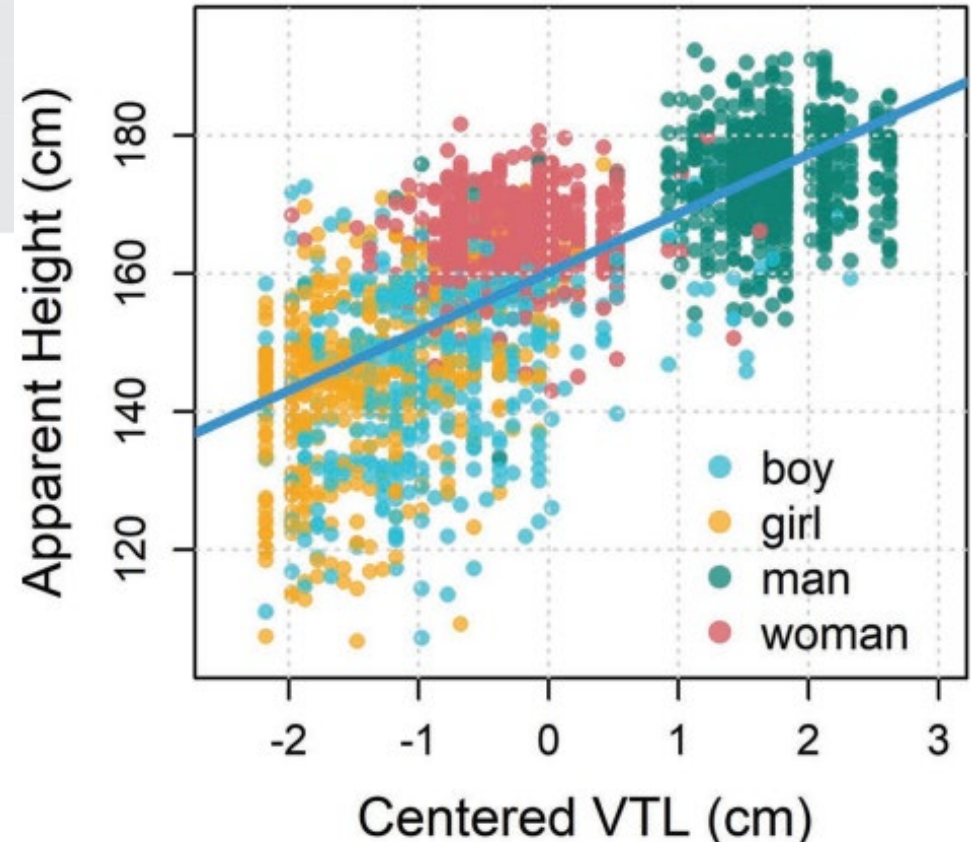
Fitting the Model

```
# Fit the model yourself
model_single_line =
  brm (height ~ vtl, data = exp_data, chains=1, cores=1,
      warmup=1000, iter = 6000,
      prior = c(set_prior("student_t(3, 160, 12)", class = "Intercept"),
                set_prior("student_t(3, 0, 12)", class = "b"),
                set_prior("student_t(3, 0, 12)", class = "sigma")))

# Or download it from the GitHub page:
model_single_line = bmb::get_model ('9_model_single_line.RDS')
```

Interpreting the Model

```
bmb::short_summary (model_single_line)
## Formula: height ~ vtl
## Population-Level Effects:
##           Estimate Est.Error 1-95% CI u-95% CI
## Intercept    160.13      0.23   159.68   160.60
## vtl           8.56      0.18    8.21    8.89
##
## Family Specific Parameters:
##           Estimate Est.Error 1-95% CI u-95% CI
## sigma      10.79      0.17   10.47   11.12
```



Models with Many Intercepts but One Slope

- Estos modelos pueden tener una intersección diferente para la línea que representa cada grupo. Sin embargo, todas las líneas tienen la misma pendiente.

```
height ~ C + vtl
```

$$\text{height}_{[i]} \sim N(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + C_{[c[i]]} + VTL \cdot \text{vtl}_{[i]}$$

Priors:

$$\text{Intercept} \sim t(3, 160, 12)$$

$$VTL \sim t(3, 0, 12)$$

$$C_{[\cdot]} \sim t(3, 0, 12)$$

$$\sigma \sim t(3, 0, 12)$$

¿Qué necesitamos para muchos valores de pendiente posibles?

Models with Many Intercepts but One Slope

- Perspectiva útil: Todos los predictores se pueden dividir en los que afectan a las pendientes y los que afectan a los interceptos.

$$\text{height}_{[i]} \sim N(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + C_{[c[i]]} + VTL \cdot \text{vtl}_{[i]}$$

Priors:

$$\text{Intercept} \sim t(3, 160, 12)$$
$$VTL \sim t(3, 0, 12)$$
$$C_{[\cdot]} \sim t(3, 0, 12)$$
$$\sigma \sim t(3, 0, 12)$$

El mismo
modelo

$$\text{height}_{[i]} \sim N(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = a_{[i]} + b_{[i]} \cdot \text{vtl}_{[i]}$$
$$a_{[i]} = \text{Intercept} + C_{[c[i]]}$$
$$b_{[i]} = VTL$$

Priors:

$$\text{Intercept} \sim t(3, 160, 12)$$
$$VTL \sim t(3, 0, 12)$$
$$C_{[\cdot]} \sim t(3, 0, 12)$$
$$\sigma \sim t(3, 0, 12)$$

Models with Many Intercepts but One Slope

- Esta parametrización deja claro por qué solo hay una pendiente, pero muchos valores de intersección posibles.

$$\text{height}_{[i]} \sim \mathcal{N}(\mu_{[i]}, \sigma)$$

$$\mu_{[i]} = a_{[i]} + b_{[i]} \cdot \text{vtl}_{[i]}$$

$$a_{[i]} = \text{Intercept} + C_{[C_{[i]}]}$$

$$b_{[i]} = VTL$$

Priors:

$$\text{Intercept} \sim t(3, 160, 12)$$

$$VTL \sim t(3, 0, 12)$$

$$C_{[\cdot]} \sim t(3, 0, 12)$$

$$\sigma \sim t(3, 0, 12)$$

Fitting the Model

```
# Fit the model yourself
model_many_intercept_one_slope =
  brm (height ~ C + vtl,
      data=exp_data, chains=1, cores=1, warmup=1000, iter=6000,
      prior = c(set_prior("student_t(3, 160, 12)", class = "Intercept"),
                set_prior("student_t(3, 0, 12)", class = "b"),
                set_prior("student_t(3, 0, 12)", class = "sigma")))

# Or download it from the GitHub page:
model_many_intercept_one_slope =
  bmmb::get_model ('9_model_many_intercept_one_slope.RDS')
```

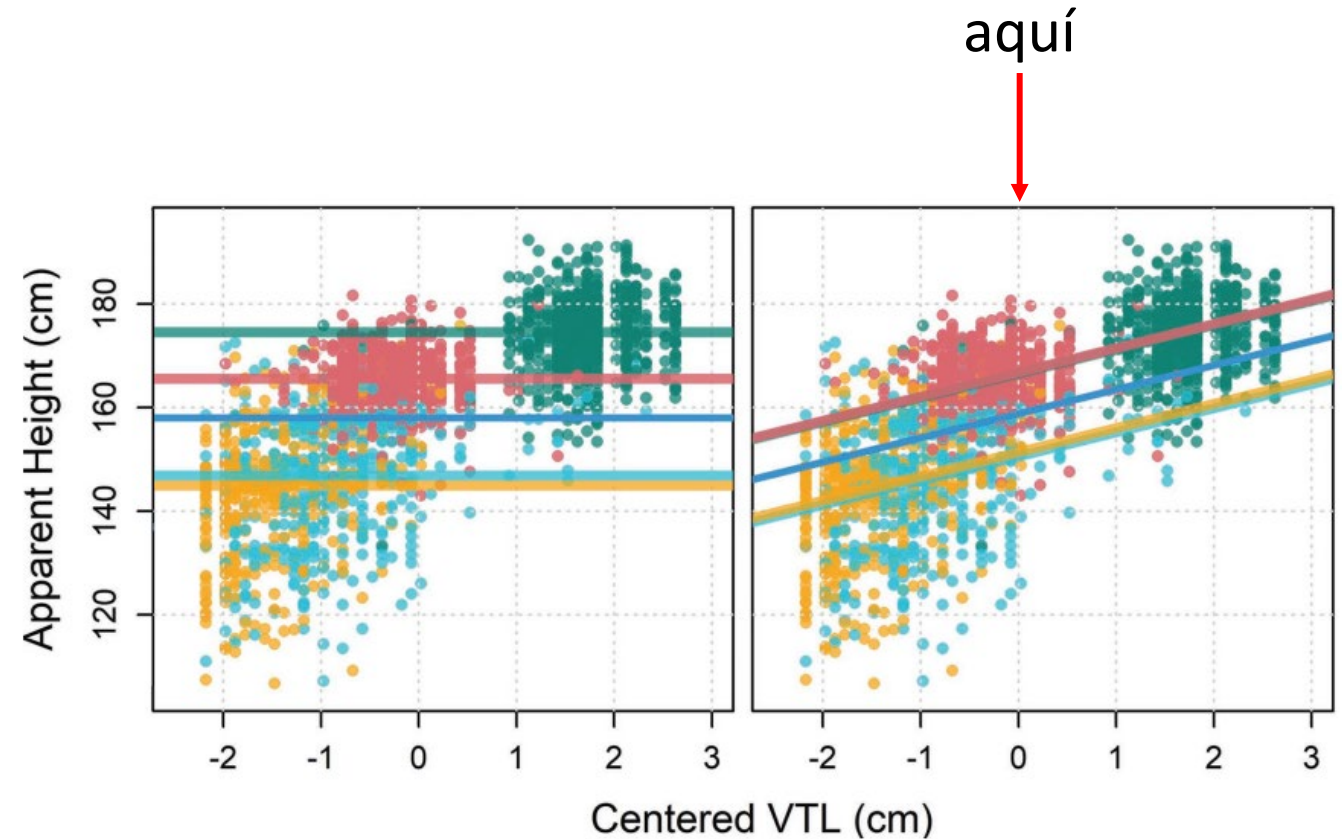
Interpreting the Model

```
many_intercept_one_slope_hypothesis = bmmmb::short_hypothesis (
  model_many_intercept_one_slope,
  hypothesis =
    c("Intercept = 0",           # overall intercept
      "Intercept + C1 = 0",      # group 1 intercept
      "Intercept + C2 = 0",      # group 2 intercept
      "Intercept + C3 = 0",      # group 3 intercept
      "Intercept + -(C1+C2+C3)=0", # group 4 intercept
      "vt1 = 0"))                # overall slope
```

```
many_intercept_one_slope_hypothesis
##      Estimate Est.Error    Q2.5    Q97.5      hypothesis
## H1   158.796     0.2138 158.374 159.214      (Intercept) = 0
## H2   150.245     0.5184 149.246 151.258      (Intercept+C1) = 0
## H3   151.503     0.6145 150.317 152.712      (Intercept+C2) = 0
## H4   166.752     0.6352 165.515 167.977      (Intercept+C3) = 0
## H5   166.685     0.3883 165.916 167.450      (Intercept+-(C1+C2+C3)) = 0
## H6    4.604     0.3169   3.974   5.219      (vt1) = 0
```

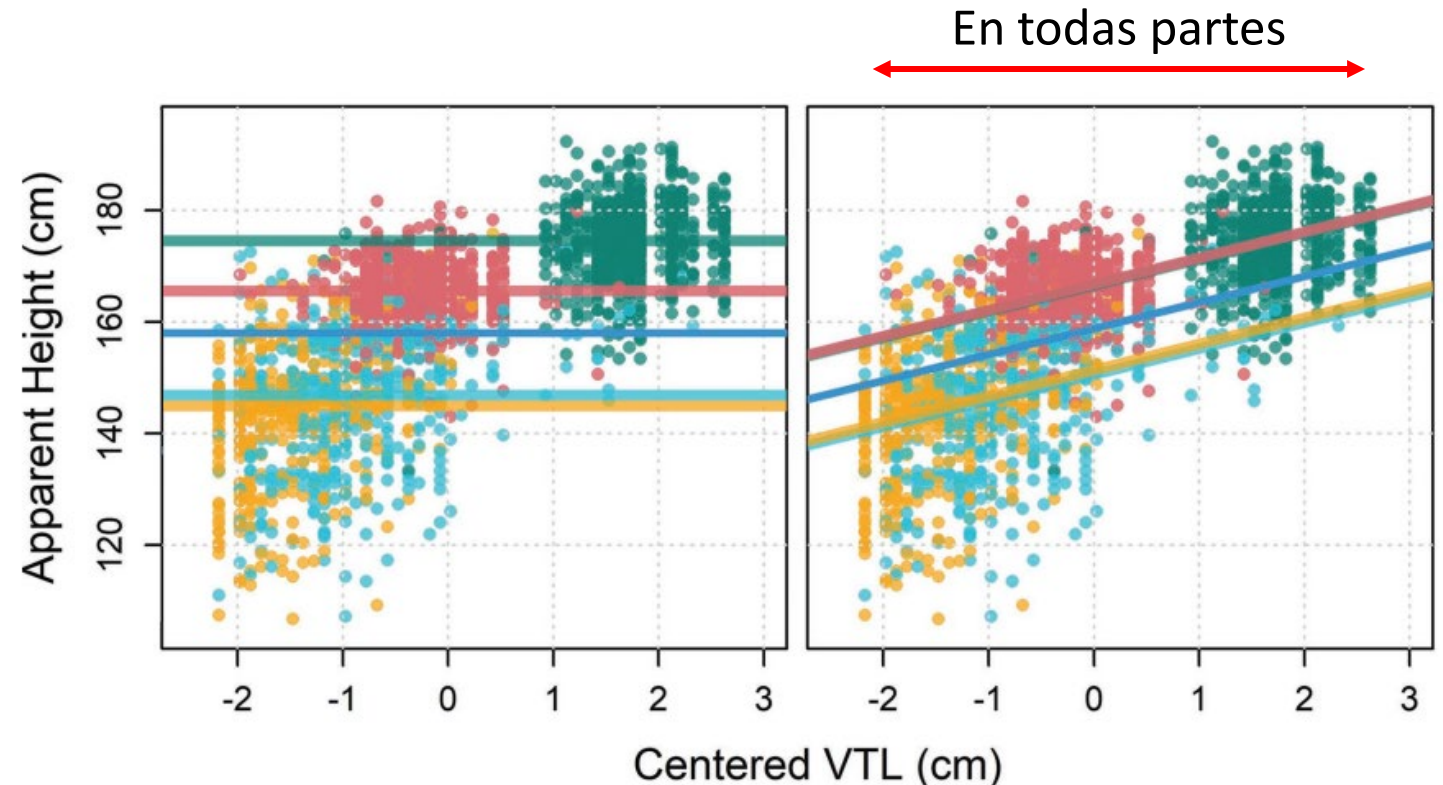
Interpreting the Model

- Cuando las líneas tienen una pendiente de cero, los efectos de grupo reflejan el espaciado de las líneas.
- Cuando los modelos tienen una pendiente distinta de cero, los efectos de grupo reflejan el espaciado de las líneas solo cuando x es exactamente cero.



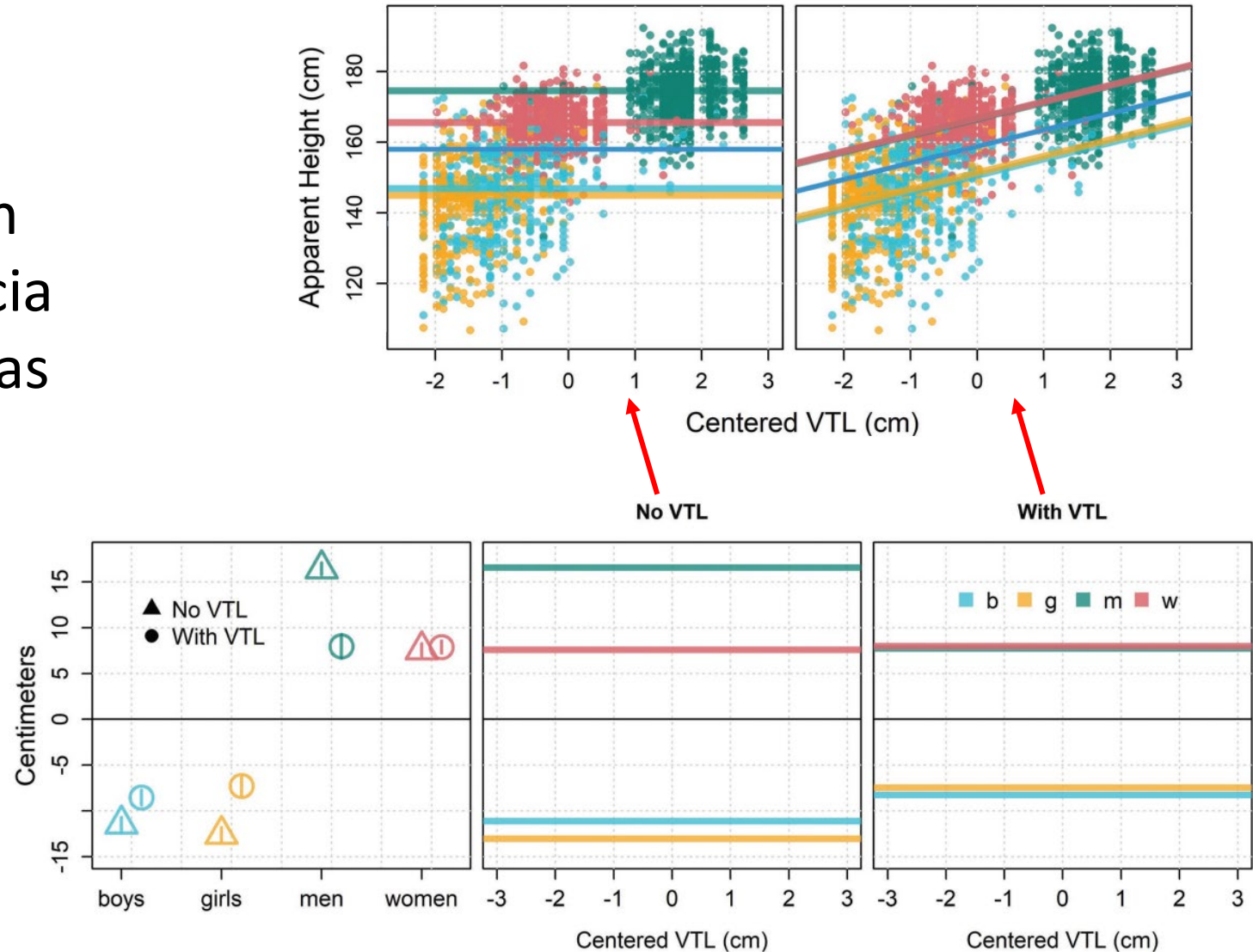
Interpreting the Model

- Sin embargo, cuando las líneas comparten una pendiente, son paralelas.
- Por lo tanto, los efectos de grupo reflejan el interlineado en todas partes.



Interpreting Group Effects with Shared Slopes

- Es probable que los efectos de grupo sean diferentes en presencia de pendientes distintas de cero.



Models with Varying Slopes and Intercepts

- Estos modelos pueden tener un intercepto y una pendiente diferente para la línea que representa cada grupo.

```
height ~ C * vtl  
height ~ C + vtl + vtl:C
```

Intercepto dependiente de
la categoría



Pendiente
"marginal"
general



Pendiente
dependiente de
la categoría



Models with Varying Slopes and Intercepts

$$\mu_{[i]} = \text{Intercept} + C_{[c[i]]} + VTL \cdot \text{vtl}_{[i]} + VTL : C_{[c[i]]} \cdot \text{vtl}_{[i]}$$

$$\mu_{[i]} = \left(\text{Intercept} + C_{[c[i]]} \right) + \left(VTL \cdot \text{vtl}_{[i]} + VTL : C_{[c[i]]} \cdot \text{vtl}_{[i]} \right)$$

$$\mu_{[i]} = \left(\text{Intercept} + C_{[c[i]]} \right) + \left(VTL + VTL : C_{[c[i]]} \right) \cdot \text{vtl}_{[i]} \quad \rightarrow \quad \begin{aligned} \mu_{[i]} &= a_{[i]} + b_{[i]} \cdot \text{vtl}_{[i]} \\ a_{[i]} &= \text{Intercept} + C_{[c[i]]} \\ b_{[i]} &= VTL + VTL : C_{[c[i]]} \end{aligned}$$

Models with Varying Slopes and Intercepts

$$\text{height}_{[i]} \sim \mathcal{N}(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \left(\text{Intercept} + C_{[c_{[i]}]} \right) + \left(VTL \cdot \text{vtl}_{[i]} + VTL : C_{[c_{[i]}]} \cdot \text{vtl}_{[i]} \right)$$

Priors:

$$\text{Intercept} \sim t(3, 160, 12)$$

$$VTL \sim t(3, 0, 12)$$

$$C_{[.]} \sim t(3, 0, 12)$$

$$VTL : C_{[.]} \sim t(3, 0, 12)$$

$$\sigma \sim t(3, 0, 12)$$

El mismo
modelo

$$\text{height}_{[i]} \sim \mathcal{N}(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = a_{[i]} + b_{[i]} \cdot \text{vtl}_{[i]}$$
$$a_{[i]} = \text{Intercept} + C_{[c_{[i]}]}$$
$$b_{[i]} = VTL + VTL : C_{[c_{[i]}]}$$

Priors:

$$\text{Intercept} \sim t(3, 160, 12)$$

$$VTL \sim t(3, 0, 12)$$

$$C_{[.]} \sim t(3, 0, 12)$$

$$VTL : C_{[.]} \sim t(3, 0, 12)$$

$$\sigma \sim t(3, 0, 12)$$

Fitting the Model

```
# Fit the model yourself
options (contrasts = c('contr.sum', 'contr.sum'))
model_multi_slope =
  brm (height ~ C + vtl + vtl:C,
      data=exp_data, chains=1, cores=1, warmup=1000, iter=6000,
      prior = c(set_prior("student_t(3, 160, 12)", class = "Intercept"),
                set_prior("student_t(3, 0, 12)", class = "b"),
                set_prior("student_t(3, 0, 12)", class = "sigma")))
```

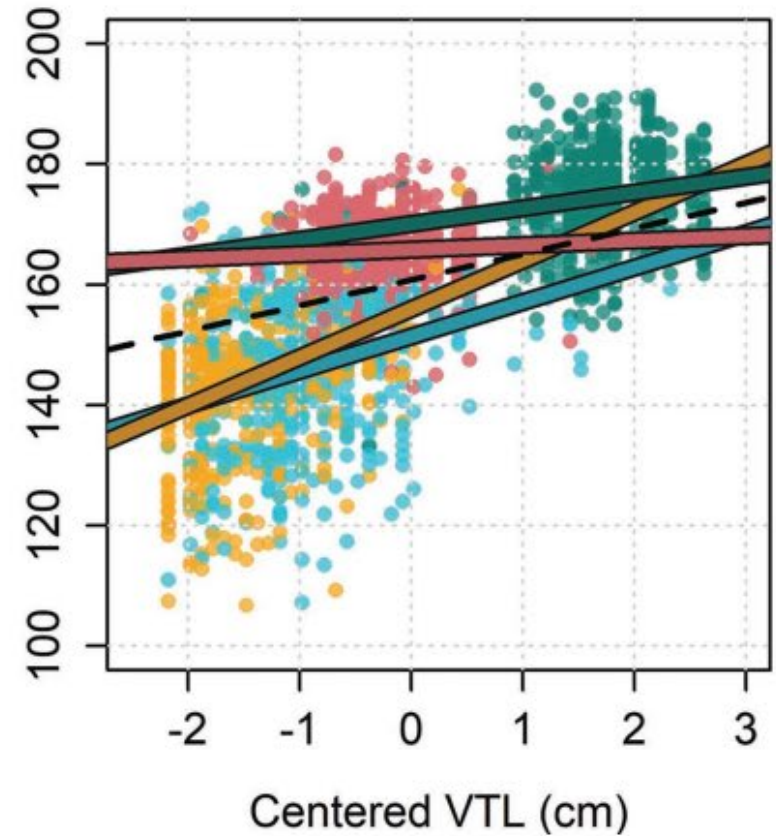
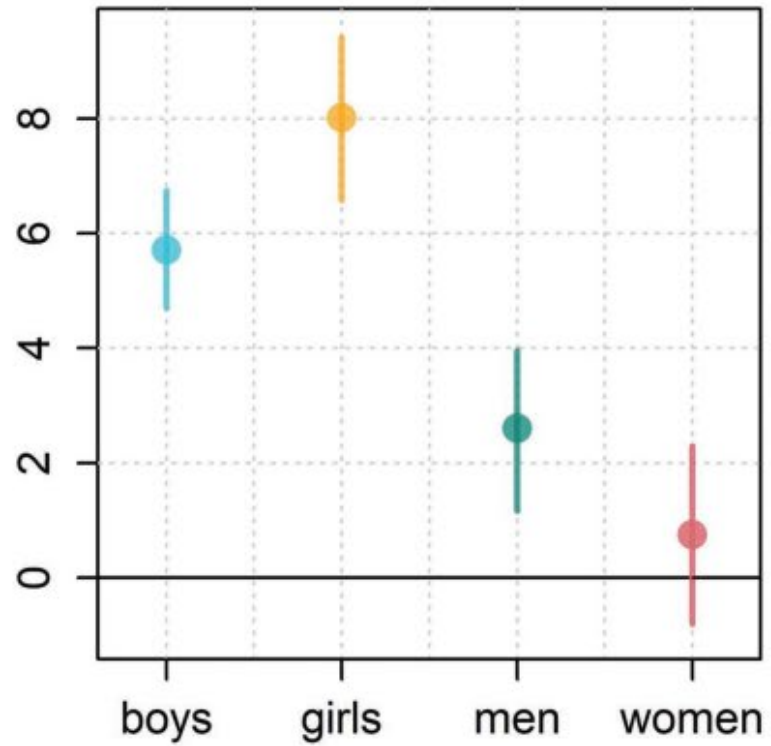
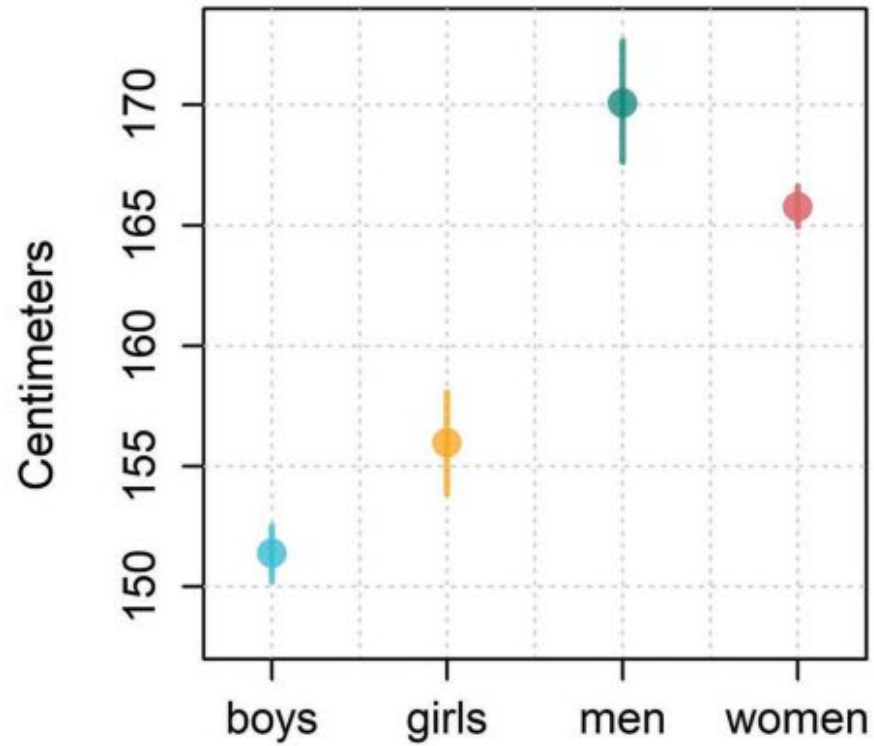
Inspecting the Fixed Effects

	<pre># inspect fixed effects</pre>	
	<pre>brms::fixef (model_multi_slope)</pre>	
	<pre>##</pre>	
Efecto principal de intercepción	<pre>## Intercept 160.800 0.4525 159.935 161.690</pre>	
Intercepciones dependientes de la categoría	<pre>## C1 -9.420 0.6208 -10.632 -8.221</pre>	
	<pre>## C2 -4.834 0.8904 -6.580 -3.090</pre>	
	<pre>## C3 9.279 1.0062 7.316 11.283</pre>	
Efecto principal de la pendiente	<pre>## vt1 4.271 0.3463 3.582 4.937</pre>	
	<pre>## C1:vt1 1.438 0.5067 0.448 2.433</pre>	
Pendientes dependientes de la categoría	<pre>## C2:vt1 3.747 0.6181 2.544 4.937</pre>	
	<pre>## C3:vt1 -1.666 0.6069 -2.913 -0.507</pre>	

Group-Specific Line Parameters

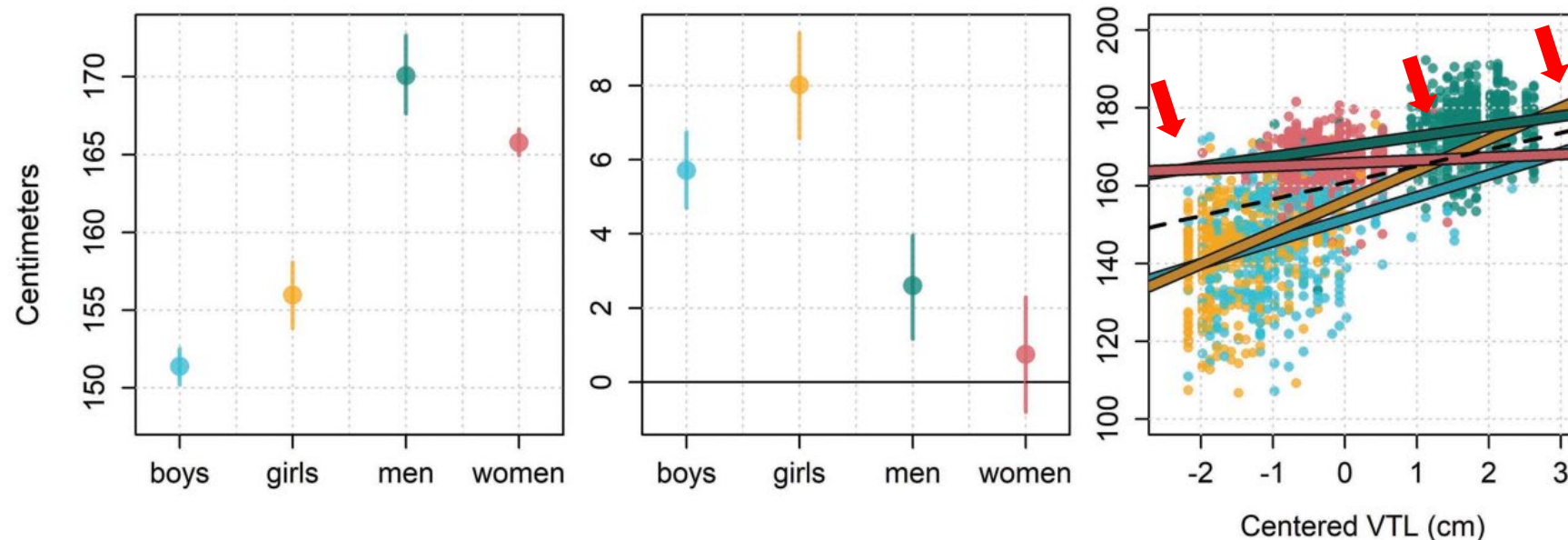
```
multi_slope_hypothesis = bmb::short_hypothesis (  
  model_multi_slope,  
  hypothesis =  
    c("Intercept = 0",                                # overall intercept  
      "Intercept + C1 = 0",                            # group 1 mean  
      "Intercept + C2 = 0",                            # group 2 mean  
      "Intercept + C3 = 0",                            # group 3 mean  
      "Intercept + -(C1+C2+C3) = 0",                  # group 4 mean  
      "vt1 = 0",                                        # overall slope  
      "vt1 + C1:vt1 = 0",                              # group 1 slope  
      "vt1 + C2:vt1 = 0",                              # group 2 slope  
      "vt1 + C3:vt1 = 0",                              # group 3 slope  
      "vt1 + -(C1:vt1+C2:vt1+C3:vt1) = 0"))           # group 4 slope
```

Group-Specific Line Parameters



Interpreting Group Effects with Varying Slopes

- Los efectos de grupo reflejan la separación de líneas solo exactamente en cero.
- La separación (es decir, los efectos de grupo) pueden ser sustancialmente diferentes en otros lugares.



Multilevel Models

- Los modelos multinivel con pendientes o intersecciones aleatorias tienen efectivamente diferentes pendientes o intersecciones para diferentes niveles del factor de agrupación (por ejemplo, oyente).

```
height ~ C + vtl + vtl:C
```

```
height ~ L + vtl + vtl:L
```

```
height ~ vtl + (vtl|L)
```

Data and Research Questions

- **L**: A number from 1 to 15 indicating which *listener* responded to the trial.
- **height**: A number representing the *height* (in centimeters) reported for the speaker on each trial.
- **S**: A number from 1 to 139 indicating which *speaker* produced the trial stimulus.
- **vtl**: An estimate of the speaker's *vocal-tract length* in centimeters.
- **G**: The *apparent gender* of the speaker indicated by the listener, **f** (female) or **m** (male).
- **A**: The *apparent age* of the speaker indicated by the listener, **a** (adult) or **c** (child).

(Q1) What is the linear relationship between speaker VTL and apparent height?

(Q2) Is the effect of VTL on apparent height affected by the apparent age and gender of the speaker?

Description of the Model

```
height ~ vtl + (vtl | L) + (1 | S)
```

$$\text{height}_{[i]} \sim t(v, \mu_{[i]}, \sigma)$$

$$\mu_{[i]} = a + b \cdot \text{vtl}_{[i]}$$

$$a_{[i]} = \text{Intercept} + L_{[L[i]]} + S_{[S[i]]}$$

$$b_{[i]} = VTL + VTL : L_{[L[i]]}$$

Priors:

$$S_{[\cdot]} \sim t(3, 0, \sigma_S)$$

$$\begin{bmatrix} L_{[\cdot]} \\ VTL : L_{[\cdot]} \end{bmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma \right)$$

$$\text{Intercept} \sim t(3, 156, 12)$$

$$VTL \sim t(3, 0, 12)$$

$$\sigma_L, \sigma_{VTL:L}, \sigma_S \sim t(3, 0, 12)$$

$$\sigma \sim t(3, 0, 12)$$

$$v \sim \text{gamma}(2, 0.1)$$

$$R \sim \text{LKJCorr}(2)$$

Fitting the Model

```
# Fit the model yourself
priors = c(brms::set_prior("student_t(3,160, 12)", class = "Intercept"),
           brms::set_prior("student_t(3,0, 12)", class = "b"),
           brms::set_prior("student_t(3,0, 12)", class = "sd"),
           brms::set_prior("lkj_corr_cholesky (2)", class = "cor"),
           brms::set_prior("gamma(2, 0.1)", class = "nu"),
           brms::set_prior("student_t(3,0, 12)", class = "sigma"))

model_random_slopes_simple =
  brms::brm (height ~ vtl + (vtl|L) + (1|S), data = exp_data, chains = 4,
            cores = 4, warmup = 1000, iter = 5000, thin = 4,
            prior = priors, family = "student")
```

Interpreting the Model

```
bmb::short_summary(model_random_slopes_simple)
## Formula: height ~ vtl + (vtl | L) + (1 | S)
##
## Group-Level Effects:
## ~L (Number of levels: 15)
##
```

	Estimate	Est.Error	l-95% CI	u-95% CI
sd(Intercept)	3.67	0.80	2.44	5.56
sd(vtl)	2.84	0.62	1.93	4.31
cor(Intercept,vtl)	-0.41	0.22	-0.76	0.07

```
##
## ~S (Number of levels: 139)
##
```

	Estimate	Est.Error	l-95% CI	u-95% CI
sd(Intercept)	4.53	0.34	3.89	5.25

```
##
## Population-Level Effects:
##
```

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	160.60	1.07	158.56	162.76
vtl	8.39	0.82	6.73	9.94

```
##
## Family Specific Parameters:
##
```

	Estimate	Est.Error	l-95% CI	u-95% CI
sigma	6.83	0.22	6.41	7.27
nu	5.24	0.73	4.05	6.82

Interpreting the Model

```
bmb::short_summary(model_random_slopes_simple)
## Formula: height ~ vtl + (vtl | L) + (1 | S)
##
## Group-Level Effects:
## ~L (Number of levels: 15)
##
##           Estimate Est.Error l-95% CI u-95% CI
## sd(Intercept)      3.67      0.80      2.44      5.56
## sd(vtl)             2.84      0.62      1.93      4.31
## cor(Intercept,vtl) -0.41      0.22     -0.76      0.07
##
## ~S (Number of levels: 139)
##
##           Estimate Est.Error l-95% CI u-95% CI
## sd(Intercept)      4.53      0.34      3.89      5.25
##
## Population-Level Effects:
##           Estimate Est.Error l-95% CI u-95% CI
## Intercept      160.60      1.07     158.56     162.76
## vtl              8.39      0.82       6.73       9.94
##
## Family Specific Parameters:
##           Estimate Est.Error l-95% CI u-95% CI
## sigma          6.83      0.22       6.41       7.27
## nu              5.24      0.73       4.05       6.82
```

```
bmb::short_summary(model_single_line)
## Formula: height ~ vtl
## Population-Level Effects:
##           Estimate Est.Error l-95% CI u-95% CI
## Intercept      160.13      0.23     159.68     160.60
## vtl              8.56      0.18       8.21       8.89
##
## Family Specific Parameters:
##           Estimate Est.Error l-95% CI u-95% CI
## sigma          10.79      0.17     10.47     11.12
```

Inspecting the Random Effects

```
# get listener effects
listener_effects = short_hypothesis(
  model_random_slopes_simple, c("Intercept = 0", "vt1 = 0"),
  scope = "ranef", group="L")
```

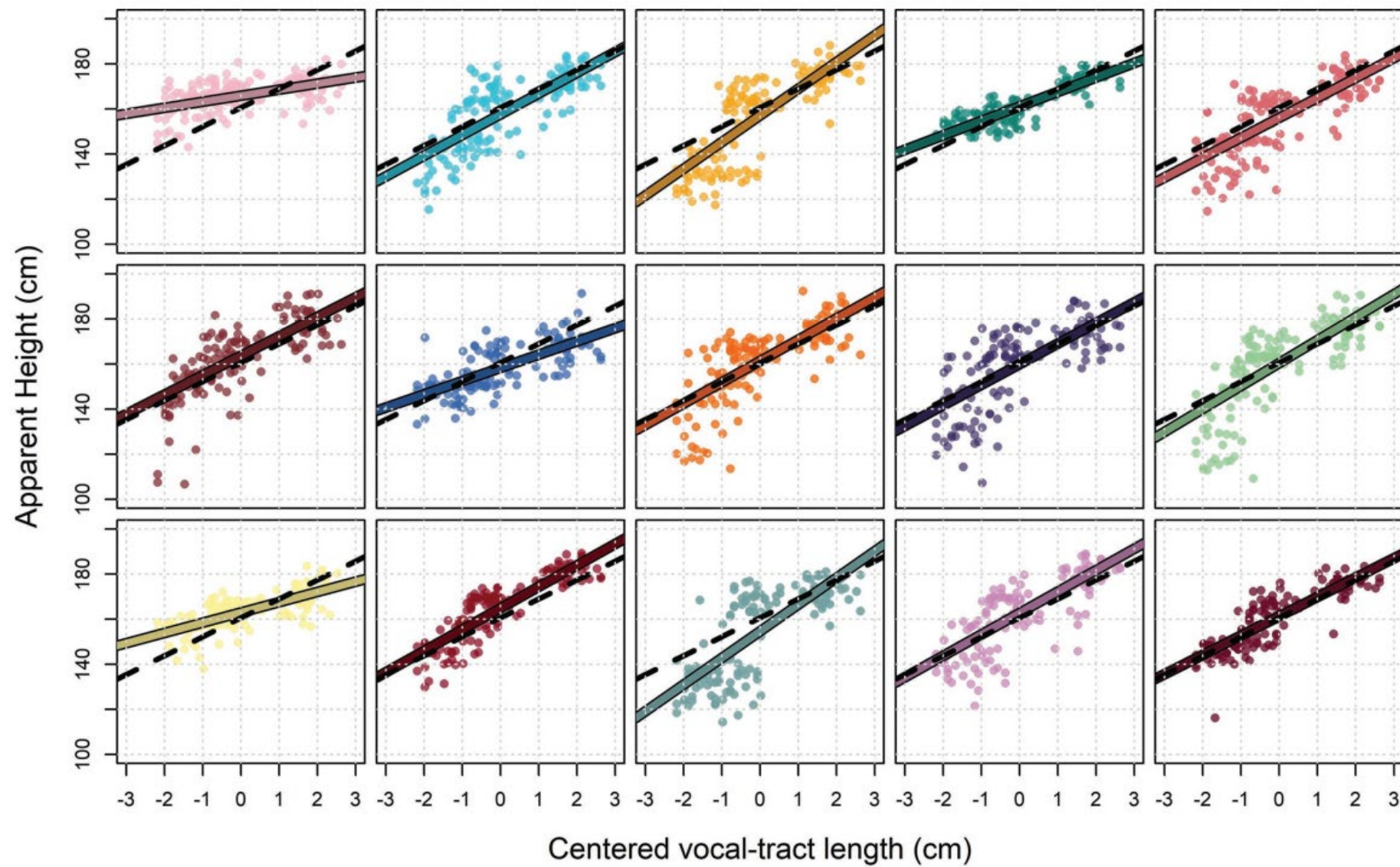
```
# get 3rd and 18th row of our listener effects
# these are the intercept and slope effects for listener 3
listener_effects[c(3,18),]
##      Estimate Est.Error   Q2.5   Q97.5      hypothesis group
## H3      -3.482    1.1836 -5.887 -1.186 (Intercept) = 0      3
## H18      3.448    0.8921  1.699  5.226   (vt1) = 0      3
```

Inspecting the Listener Coefficients

```
# get listener coefficients
listener_coefficients = short_hypothesis(
  model_random_slopes_simple, c("Intercept = 0", "vtl = 0"),
  scope = "coef", group="L")
```

```
# get 3rd and 18th row of our listener coefficients
# these are the intercept and slope coefficients for listener 3
listener_coefficients[c(3,18),]
##      Estimate Est.Error   Q2.5   Q97.5      hypothesis group
## H3      157.12    0.7981 155.56 158.73 (Intercept) = 0      3
## H18      11.84    0.5667  10.71  12.95   (vtl) = 0      3
```

Inspecting the Listener Coefficients



A Model with 'Random Slopes'

```
height ~ vtl*A*G + (vtl*A*G|L) + (1|S)
```

```
height ~ vtl + A + G + A:G + vtl:A + vtl:G + vtl:A:G +  
          (vtl + A + G + A:G + vtl:A + vtl:G + vtl:A:G|L) + (1|S)
```

Description of the Model

$$\text{height}_{[i]} \sim t(v, \mu_{[i]}, \sigma)$$

$$\mu_{[i]} = a + b \cdot \text{vtl}_{[i]}$$

$$a_{[i]} = \text{Intercept} + A + G + A : G + \\ A : L_{[L[i]]} + G : L_{[L[i]]} + A : G : L_{[L[i]]} + L_{[L[i]]} + S_{[S[i]]}$$

$$b_{[i]} = VTL + VTL : A + VTL : G + VTL : A : G + \\ VTL : A : L_{[L[i]]} + VTL : G : L_{[L[i]]} + VTL : A : G : L_{[L[i]]} + VTL : L_{[L[i]]}$$

Priors :

$$S_{[\bullet]} \sim t(3, 0, \sigma_S)$$

$$\begin{bmatrix} A : L_{[\bullet]} \\ G : L_{[\bullet]} \\ A : G : L_{[\bullet]} \\ L_{[\bullet]} \\ VTL : A : L_{[\bullet]} \\ VTL : G : L_{[\bullet]} \\ VTL : A : G : L_{[\bullet]} \\ VTL : L_{[\bullet]} \end{bmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \Sigma \right)$$

Fitting the Model

```
# Fit the model yourself
priors = c(brms::set_prior("student_t(3,160, 12)", class = "Intercept"),
          brms::set_prior("student_t(3,0, 12)", class = "b"),
          brms::set_prior("student_t(3,0, 12)", class = "sd"),
          brms::set_prior("lkj_corr_cholesky (2)", class = "cor"),
          brms::set_prior("gamma(2, 0.1)", class = "nu"),
          brms::set_prior("student_t(3,0, 12)", class = "sigma"))

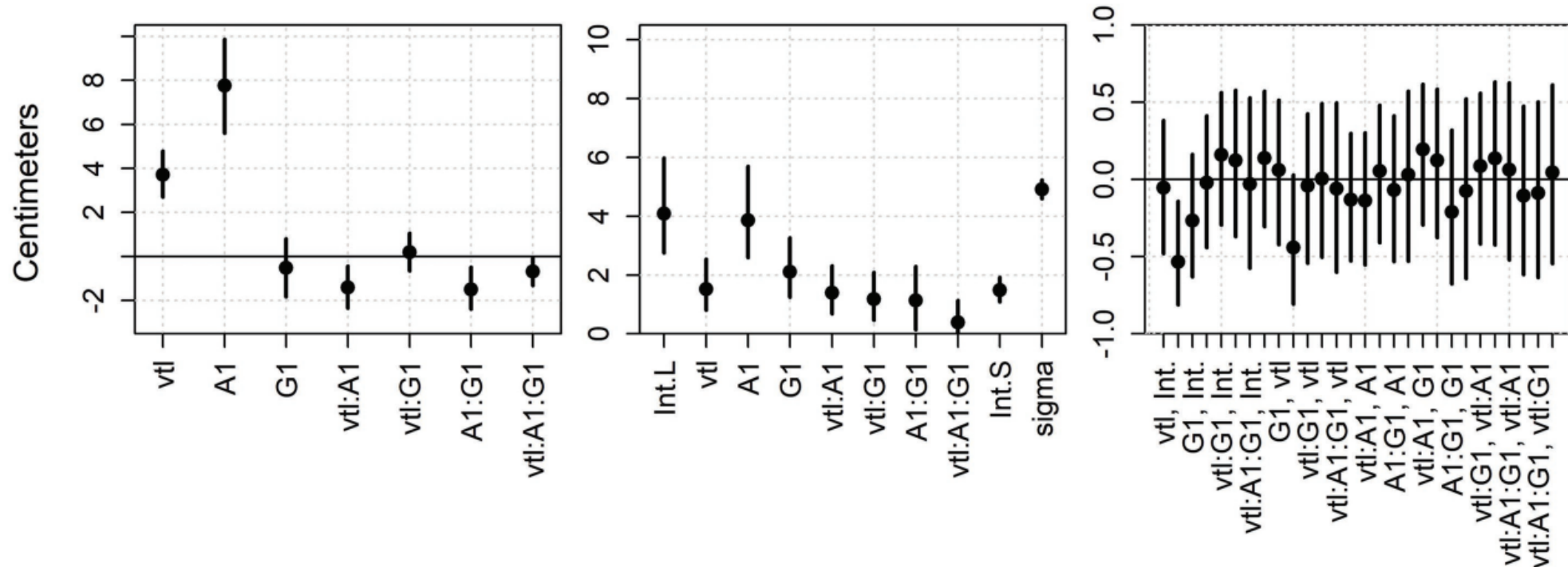
model_random_slopes_complex =
  brms::brm (height ~ vtl*A*G + (vtl*A*G|L) + (1|S), data = exp_data,
            chains = 4, cores = 4, warmup = 1000, iter = 5000, thin = 4,
            prior = priors, family = "student")
```

Interpreting the Model

```
# model fixed effects
fixef_effects = brms::fixef (model_random_slopes_complex)

# model random effect standard deviations
sds = bmb::get_sds (model_random_slopes_complex)

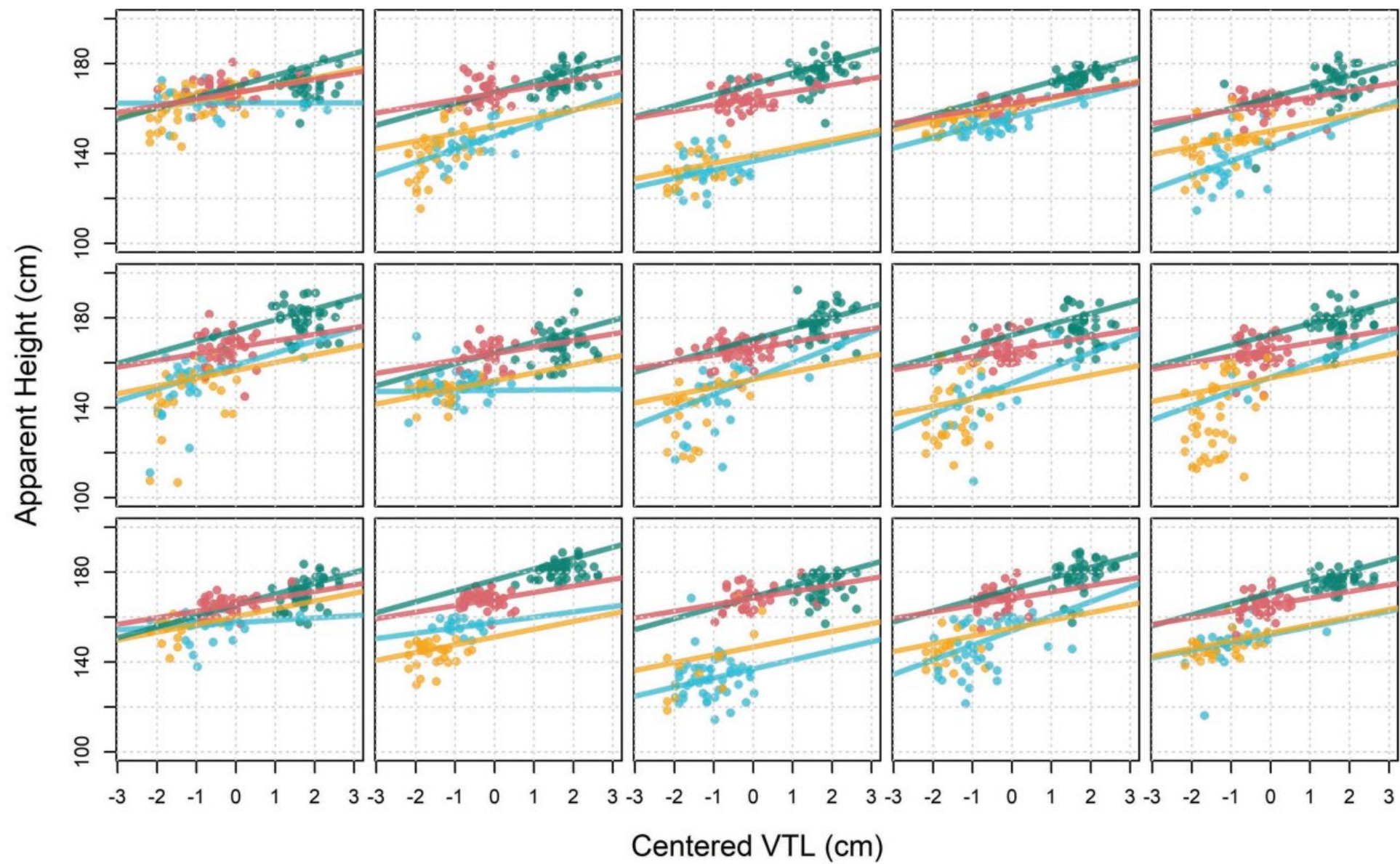
# model random effect correlations
correlations = bmb::get_corrs (model_random_slopes_complex, factor="L")
```



Listener and Group-Specific Parameters

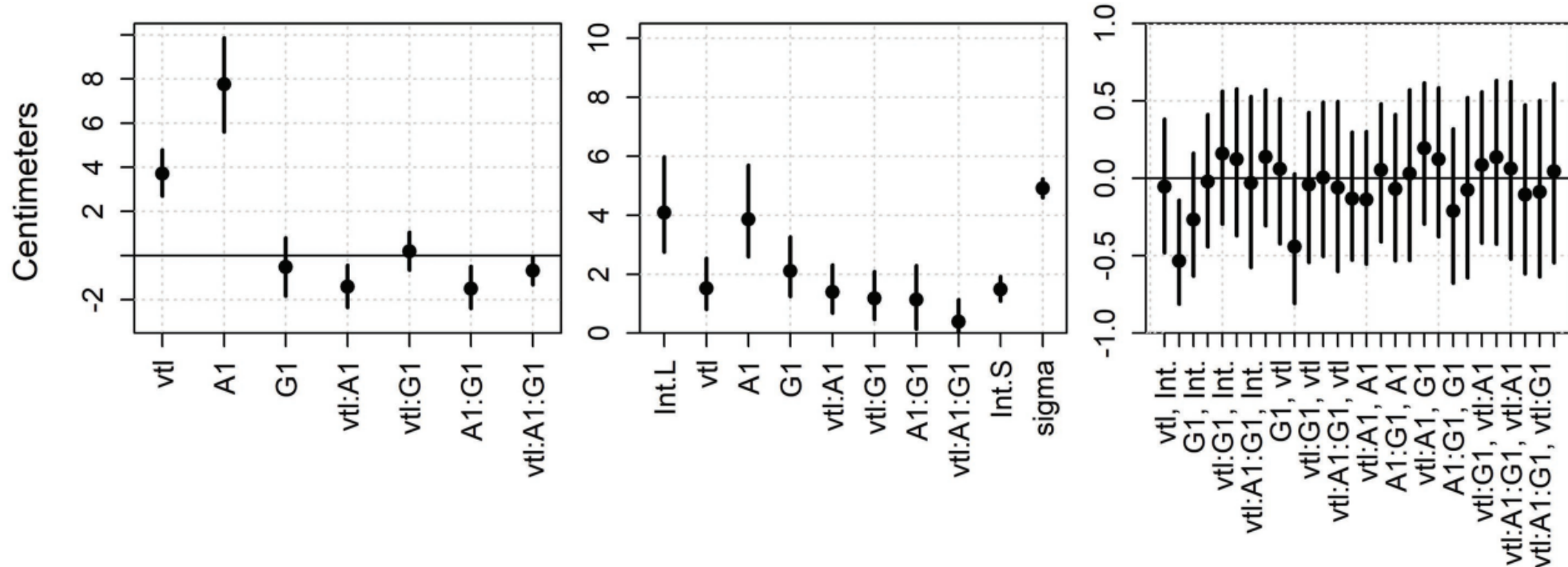
```
random_slopes_complex_hypothesis_listener = bmbb::short_hypothesis (
  model_random_slopes_complex,
  hypothesis =
    c("Intercept - A1 - G1 + A1:G1 = 0",          # group 1 mean
      "Intercept - A1 + G1 - A1:G1 = 0",          # group 2 mean
      "Intercept + A1 - G1 - A1:G1 = 0",          # group 3 mean
      "Intercept + A1 + G1 + A1:G1 = 0",          # group 4 mean
      "vt1 - vt1:A1 - vt1:G1 + vt1:A1:G1 = 0",    # group 1 slope
      "vt1 - vt1:A1 + vt1:G1 - vt1:A1:G1 = 0",    # group 2 slope
      "vt1 + vt1:A1 - vt1:G1 - vt1:A1:G1 = 0",    # group 3 slope
      "vt1 + vt1:A1 + vt1:G1 + vt1:A1:G1 = 0"),    # group 4 slope
  scope = "coef", group="L")
```

Listener and Group-Specific Parameters



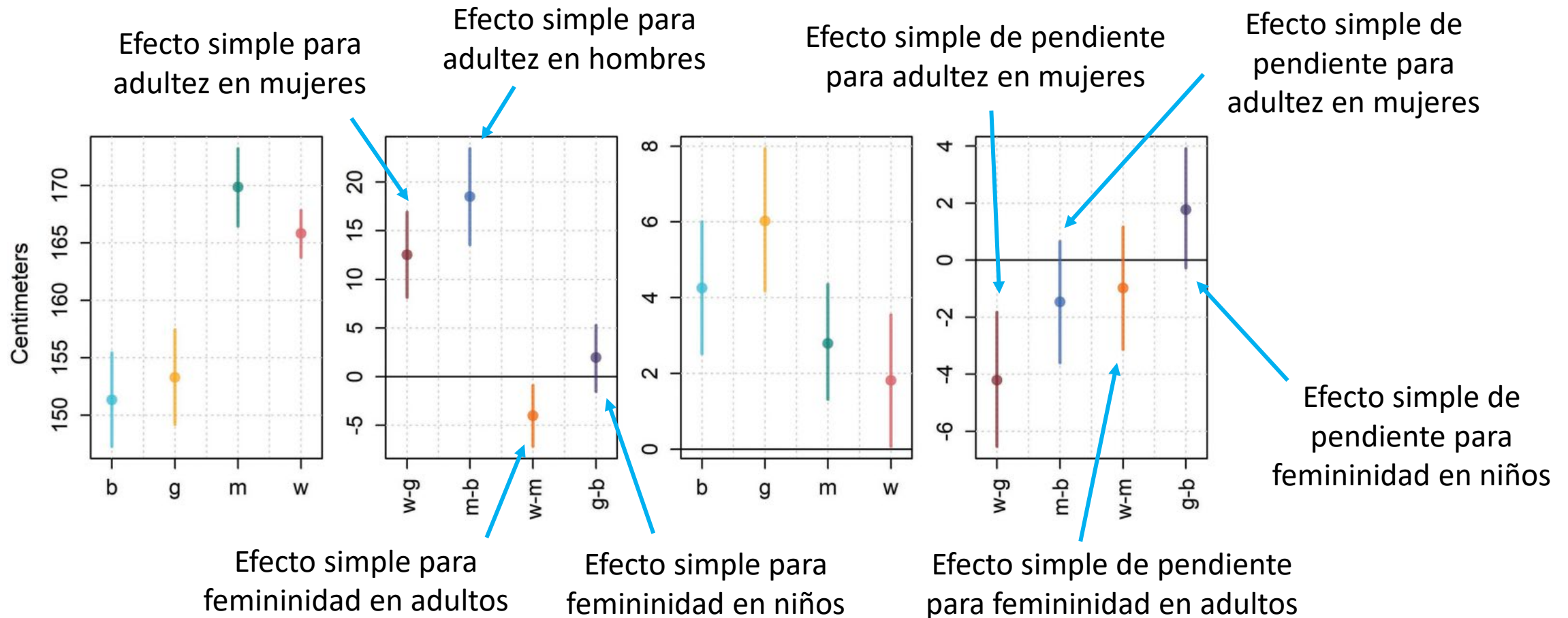
Answering our Research Questions

- No: Listes de un montón de números.
- Lo que hay que hacer: Cuéntale al lector la historia de los números, qué significan todos ellos.



Answering our Research Questions

- Encuentre pendientes e interceptos específicos de un grupo e inspeccione los efectos simples.



Answering our Research Questions

```
simple_effects = bmb::short_hypothesis (  
  model_random_slopes_complex,  
  hypothesis =  
    c("2*(A1 + A1:G1) = 0",          # age difference for women  
      "2*(A1 - A1:G1) = 0",          # age difference for men  
      "2*(G1 + A1:G1) = 0",          # gender difference for adults  
      "2*(G1 - A1:G1) = 0",          # gender difference for children  
      "2*(vtl:A1 + vtl:A1:G1) = 0",  # VTL difference by age for women  
      "2*(vtl:A1 - vtl:A1:G1) = 0",  # VTL difference by age for men  
      "2*(vtl:G1 + vtl:A1:G1) = 0",  # VTL difference by gender for adults  
      "2*(vtl:G1 - vtl:A1:G1) = 0")) # VTL difference by gender for children
```

Causality

- Difícil, si no imposible, "probar" que algo causa otra cosa.
- Muchos ejemplos de inferencias erróneas con respecto a la causalidad. Imposible decir en qué nos equivocaremos en el futuro.
- Nuestros modelos pueden ayudarnos a entender las relaciones, pero no las causas o los mecanismos exactos.

Exercises: Week 8

Use the data in 'exp_ex' or use your own data to answer a related question. In any case, describe the model, present and explain the results. More requirements:

- You must include a model at least this complex: $y \sim A*B + (A*B|L)$, where either A or B is a quantitative predictor.
- Fit a new model.
- Include at least two non-book figures.
- Include a table of your fixed effects.
- Report and interpret at least one simple effect.